PROJECT SPECIFICATIONS

Multi-Modal Agentic AI System for Medical Diagnosis and Triage

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Team GitHub Repo: https://github.com/abhishek7112000/Multi-Modal-Agentic-AI-System-for-Medical-Diagnosis-and-Triage

1. Project Topic

This project develops a Multi-Modal Agentic AI System for Medical Diagnosis and Triage. The system addresses a real-world clinical problem: enabling faster, more consistent preliminary diagnosis and triage for chest and cardiovascular conditions, reducing unnecessary A&E attendance and GP delays. The solution uses machine learning across three modalities — chest X-ray imaging, structured electronic health records (EHR), and clinical free text — and integrates them within a stateful AI agent (LangGraph.js) backed by MedGemma-27B-IT as the reasoning LLM. A patient-facing Next.js application serves as the front end, providing a diagnosis report with triage recommendation and automated appointment booking.

2. Research Question / Problem Statement

Primary Research Question:

> *Can a multi-modal machine learning pipeline — combining chest X-ray classification, structured EHR risk scoring, and clinical NER — produce reliable preliminary diagnosis and triage decisions for chest and cardiovascular conditions, and can this be deployed as a real-time, user-facing application?*

Problem Statement:

Delayed or missed triage of chest and cardiovascular symptoms (AMI, heart failure, pneumonia, COPD, AFib) is a leading cause of preventable harm in primary care. Existing tools are unimodal and cannot synthesise imaging, labs, and clinical text simultaneously. This project addresses that gap by training three complementary ML models — a custom neural network on NIH ChestX-ray14 images, a hybrid XGBoost+NER model on Synthea synthetic EHR data, and a fine-tuned biomedical NER model on Synthea encounter notes — and orchestrating them within an AI agent to produce a structured, explainable DiagnosisReport with a four-level triage decision (EMERGENCY / URGENT / ROUTINE / SELF_CARE).

3. Objectives

- Acquire, pre-process, and split two real-world datasets: NIH ChestX-ray14 (112,120 X-ray images) and Synthea Synthetic EHR (patient records, labs, and encounter notes).
- Fine-tune and compare three Model 1 (ChestVision) architectures using transfer learning via timm — ResNet50V2, ViT-B/16, and EfficientNetV2B0 — for multi-label classification of 14 thoracic conditions from chest X-rays. ViT-B/16 selected as final deployed model based on highest validation AUC. (Note: ViT-B/16 from scratch was discarded — dataset of ~100k images is insufficient for scratch training of vision transformers.)
- Train Model 2 (ClinicalFusion) — a hybrid MLP tabular branch + frozen BioClinical-ModernBERTlarge text branch for multi-label condition prediction and severity scoring (0–3) from structured EHR data.
- Fine-tune Model 3 (OpenMed NER) — transfer learning on OpenMed-NER-DiseaseDetectSuperClinical-434M to extract chest/cardiovascular entities from Synthea encounter notes.
- Apply hyperparameter optimisation (Optuna Bayesian search for Models 1 & 2; layer-freezing strategy for Model 3) to maximise performance against defined KPIs.

- Export all three trained models to ONNX format with INT8 quantisation; deploy via onnxruntimenode within Vercel serverless functions (250 MB limit).
- Build a LangGraph.js stateful AI agent orchestrating the three models and MedGemma-27B-IT to produce a final DiagnosisReport enriched by RAG (A-Z Medical Encyclopedia via ChromaDB).
- Produce a full-stack Next.js 14 patient-facing application with Clerk authentication, NeonDB, Google Calendar/Gmail appointment booking, and real-time SSE agent progress display.

4. Goal / Target

The overarching goal is to deliver a deployable, evidence-based AI triage assistant that processes multi-modal patient data and returns a clinically grounded, explainable diagnosis report — demonstrating the integration of all three required ML model types within a production-grade application.

Target Area	Description	Success Measure
Model 1 — ChestVision	Fine-tuned ViT-B/16 (selected from comparison of ResNet50V2, ViTB/16, EfficientNetV2B0) classifying 14 thoracic conditions from X-rays via transfer learning	Mean AUC-ROC ≥ 0.82
Model 2 — ClinicalFusion	Hybrid MLP tabular branch + frozen BioClinical-ModernBERT-large text branch.	Macro F1 ≥ 0.80; False low rate < 2%
Model 3 — OpenMed NER	Fine-tuned pre-trained NER model extracting chest/cardiac entities from encounter notes	Entity F1 ≥ 0.82; Transfer gain ≥ 3 F1 pts
Optimisation	Hyperparameter tuning via Optuna (50 trials) for Models 1 & 2; differential LR fine-tuning for Model	≥5% improvement over untuned baseline
ONNX Deployment	3 All three models exported, quantised (INT8), and running in Vercel serverless (≤250 MB each)	CPU latency: <50 ms (M1), <20 ms (M2), <30 ms (M3)
Application	Next.js 14 app with LangGraph.js agent, MedGemma-27B-IT, Clerk, NeonDB, Calendar/Gmail	UAT passed; full E2E smoke test with synthetic patient
GitHub Documentation	All 6 project plans committed to week 3 repo; README and file naming system in place	All plans present and correctly formatted on GitHub by deadline

5. Key Performance Indicators (KPIs) — Defining Success

5.1 Model 1 — Chest Vision (Custom Neural Network)

- KPI 1: Three architectures compared: ResNet50V2, ViT-B/16 (selected), EfficientNetV2B0. All fine-tuned via two-phase transfer learning using timm pretrained ImageNet weights.
- KPI 2: Per-class AUC-ROC ≥ 0.75 for each of the 14 pathologies.
- KPI 3: F1 score ≥ 0.60 for the top 5 clinically relevant conditions at Youden's J optimal threshold.
- KPI 4: ONNX inference latency < 50 ms per image on CPU (onnxruntime-node / Vercel serverless).

5.2 Model 2 — Clinical Fusion (Hybrid Model)

- KPI 5: Macro F1 (diagnosis, 5 classes) ≥ 0.80; Weighted F1 ≥ 0.85.
- KPI 6: ROC-AUC (One-vs-Rest) ≥ 0.90 per diagnosis category.
- KPI 7: Severity MAE (ordinal 0–3) < 0.4; Cohen's Kappa ≥ 0.75.
- KPI 8: False low rate (critically unwell patients misclassified as low severity) < 2% on held-out test set. This is the primary clinical safety metric.
- KPI 9: ONNX inference latency < 20 ms per patient record on CPU.

5.3 Model 3 — OpenMed NER (Pre-Trained + Transfer Learning)

- KPI 10: Entity-level F1 (seqeval) ≥ 0.82 overall; ≥ 0.80 per entity type (DISEASE, ANATOMY, PHARMA).
- KPI 11: DISEASE recall ≥ 0.85 (safety-critical: missing a disease mention is worse than a false positive).
- KPI 12: Transfer gain — fine-tuned model must exceed the base OpenMed model by ≥ 3 F1 points on the Synthea chest/cardiac test set.
- KPI 13: ONNX entity F1 parity within 0.5 percentage points of the PyTorch model.

5.4 Optimization KPIs

- KPI 14: Optuna hyper parameter search (50 trials) achieves ≥ 5% AUC improvement over default configuration for Model 1.
- KPI 15: Clinical Fusion full fusion (XGBoost + OpenMed NER encoder) outperforms XGBoost-only ablation by ≥ 5 F1 points, demonstrating measurable benefit of the hybrid architecture.

5.5 Application & GitHub KPIs

- KPI 16: All three ONNX models deployed and returning inference results within the Vercel 250 MB function size limit.
- KPI 17: LangGraph.js agent completes full E2E diagnostic workflow (intake → NER → X-ray → clinical → MedGemma → RAG → triage) within 120 seconds for a standard case.
- KPI 18: All six project plans committed to the week 3 GitHub repository before the submission deadline, with consistent file naming and a current README.
- KPI 19: User Acceptance Test (UAT) completed with no critical failures using a synthetic patient scenario.

5.6 Clinical Safety KPIs

- KPI 20: Zero under-triage of EMERGENCY cases in UAT — the agent must never output ROUTINE or SELF_CARE for any synthetic case flagged as EMERGENCY in the reference set. This is the primary system-level safety gate.
- KPI 21: End-to-end triage agreement rate ≥ 80% between agent output and a clinician-reviewed reference set of 20 synthetic cases (EMERGENCY / URGENT / ROUTINE / SELF_CARE level match), validating the full multi-modal pipeline as a coherent system.

The project is considered successful when all three models meet their KPI thresholds, the application passes UAT and all six project plans are committed to the team GitHub repository by the Week 3 deadline.